



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE (Advanced)

SCORE - II

06-06-2021

JEE(Main + Advance) : Leader & Enthusiast Course (PHASE : SCORE-II)

Test Type : SCORE - II

Test Pattern : JEE (Advanced)

TEST DATE : 06-06-2021

ANSWER KEY

PAPER-1

PHYSICS

SEC-1	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	A	C	B	B	B	A	A	A	B
SEC-2	Q.	1	2	3	4	5					
	A.	B,C	A,C,D	A,D	A,C,D	A,D					
SEC-3	Q.	1	2	3	4	5					
	A.	3	8	5	2	6					

CHEMISTRY

SEC-1	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	B	B	B	A	D	D	C	D	B
SEC-2	Q.	1	2	3	4	5					
	A.	A,B,D	A,B,D	A,B,C,D	A,B,D	A,B,C,D					
SEC-3	Q.	1	2	3	4	5					
	A.	8	4	1	9	6					

MATHEMATICS

SEC-1	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	D	B	D	A	B	C	B	A	A
SEC-2	Q.	1	2	3	4	5					
	A.	A,B,C	A,B,C,D	A,B,C	A,B,D	A,B					
SEC-3	Q.	1	2	3	4	5					
	A.	8	3	5	3	6					

HINT – SHEET

PHYSICS

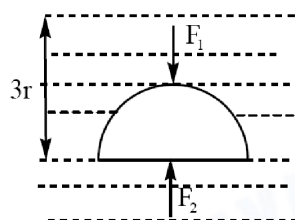
SEC-1

1. Ans (D)

Check for dimensionally correct option.

2. Ans (A)

Imagine that lower half of sphere is removed and upper hemisphere is at the given position.

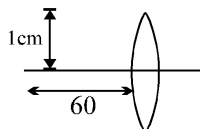


$$F_2 - F_1 = F_b$$

$$[P_0 + 3r\rho g] \pi r^2 - F_1 = \frac{2}{3} \pi r^3 \rho g$$

$$F_1 = \frac{\pi r^3}{3} [3P_0 + 7r\rho g]$$

3. **Ans (C)**



$$h_I = mh_0$$

$$= \frac{f h_0}{u + f} = \frac{40 h_0}{-60 + 40} = -2h_0$$

$$\frac{dh_I}{dt} = -2 \frac{dh_0}{dt} = -2 \times 10 = -20 \text{ cm/s}$$

4. **Ans (B)**

$$r = (K_1)n^2$$

$$T \propto r^{3/2} \Rightarrow T \propto n^3 \Rightarrow T = K_2 n^3$$

$$B = \frac{\mu_0 I}{2r} = \frac{\mu_0}{2r} \cdot \frac{e}{T} \Rightarrow B \propto \frac{1}{n^5}$$

$$M = IA = \frac{e}{T} \pi r^2 \Rightarrow M \propto n$$

$$r \propto n^2, v \propto \frac{1}{n}$$

$$\therefore \frac{Br^2}{Mv^2} \text{ will be independent of } n.$$

5. **Ans (B)**

As temperature of second body is more, its average specific heat is more. In the formula $\Delta Q = ms_{\text{ave}} (\Delta T)$, ΔQ and m are same for both. ΔT is less for 2nd one.

$$(80 - T_f) < (T_f - 20)$$

$$100 < 2T_f \Rightarrow T_f > 50^\circ\text{C}$$

6. **Ans (B)**

As the electron approaches the plate, negative potential produced by it at the plate increases gradually. To make its potential zero, earth sends +ve charge to the plate i.e. current is +ve. Reverse thing happens when electron moves away from plate. When the electron is in the middle of plate, potential of plate remains constant due to e^- and hence $I = 0$

7. **Ans (A)**

Current in resistance R is

$$\left(\frac{E}{r + \frac{50R}{50+R}} \right) \times \frac{50}{(50+R)}$$

$$\text{Power} = \left[\left(\frac{E}{50R + 50r + Rr} \right) \times 50 \right]^2 \times R$$

For maximum power

$$\frac{dP}{dR} = 0 \Rightarrow R = \frac{50}{11} \Omega$$

8. **Ans (A)**

$$E_i = \frac{k(2Q)}{(2R)^2} - \frac{k(Q)}{(4R)^2} = \frac{7kQ}{16R^2}$$

9. **Ans (A)**

From the graph $U = 3 \sin \pi x + 3$

$$\therefore F = -\frac{\partial U}{\partial x} = -3\pi \cos \pi x$$

$$\therefore F_{\text{max}} = 3\pi$$

10. **Ans (B)**

$$E_0 = \frac{2\lambda}{2\pi \epsilon_0 d}$$

$$\lambda = \pi \epsilon_0 d E_0$$

For $r \gg \ell$, charge on the rod can be treated as a point charge.

$$E = \frac{1}{4\pi \epsilon_0} \frac{q}{r^2} \text{ where } q = \lambda \ell = \pi \epsilon_0 d E_0 \ell$$

PHYSICS

SEC-2

1. **Ans (B,C)**



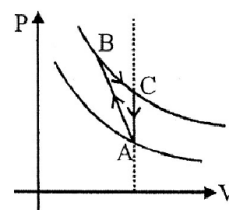
$$Mv = 2Mv^1 \rightarrow v^1 = \frac{v}{2}$$

$$2 \left(\frac{ML^2}{12} + M \left(\frac{L}{2\sqrt{2}} \right)^2 \right) \omega = Mv \frac{L}{4}$$

2. **Ans (A,C,D)**

Let us consider a closed process ABCA. CA is an isochoric process. In CA, T decreases as 'P' decreases.

$\Delta Q_{CA} = nC_V \Delta T$ is negative. Let it is $-y$.



$Q_{ABCA} = W_{ABCA}$ is +ve. Let it is $+z$

$$\Rightarrow Q_{A \rightarrow B} \Rightarrow Q_{B \rightarrow C} \Rightarrow Q_{C \rightarrow A} = +z$$

$$\Rightarrow Q_{A \rightarrow B} + 0 + (-y) = +z$$

$$\Rightarrow Q_{A \rightarrow B} = z + y > 0$$

3. **Ans (A,D)**

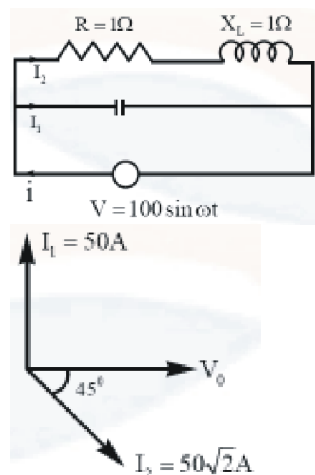
Conceptual

4. **Ans (A,C,D)**

$$i_2 = \frac{V_0}{\sqrt{R^2 + X_L^2}} = 50\sqrt{2} \text{ A and } i_1 = 50 \text{ A}$$

From phasor diagram, resultant of I_1 and I_2 will be along V_0 and $I_{\text{res}} = 50 \text{ A}$

Phase difference between v and i is $\theta = 0^\circ$

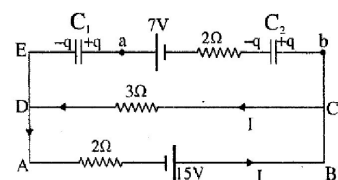


$$P_{\text{av}} = \frac{i_0 V_0}{2} \cos \theta = 2500 \text{ W}$$

$$Z = \frac{V_0}{i_0} = 2 \Omega$$

5. **Ans (A,D)**

At steady state $I(3) + I(2) = 15 \Rightarrow I = 3$



KVL $C \rightarrow D \rightarrow E \rightarrow a \rightarrow b \rightarrow c$

$$V_C - I(3) + \frac{q}{11} - 7 + \frac{q}{5} = V_C$$

$$\frac{q}{11} + \frac{q}{5} = 7 + 3 \times 3 = 16 \Rightarrow q = 55 \mu\text{C}$$

Now KVL for $a \rightarrow b$

$$V_a - 7 + \frac{q}{5} = V_b$$

$$\Rightarrow V_a - V_b = 7 - \frac{q}{5} = 7 - \frac{55}{5} = \frac{55}{5} = -4 \text{ V}$$

Potential difference across

$$C_1 : \frac{q}{11} = \frac{55}{11} = 5 \text{ V}$$

$$\Rightarrow \text{P.d. across } C_2 : \frac{q}{5} = 11 \text{ V}$$

Potential difference across terminal = $15 - I(2)$

$$= 15 - 3 \times 2 = 9 \text{ V}$$

PHYSICS

SEC-3

1. **Ans (3)**

Let initial and final speed of stone be u and v .

$$\therefore v^2 = u^2 - 2gh$$

$$\text{And } v \cos 30^\circ = u \cos 60^\circ$$

Solving above equations we get

$$u = \sqrt{3gh}$$

2. **Ans (8)**

Conserving angular momentum $m(v_1 \cos 60^\circ)$

$$4R = mv_2 R \Rightarrow \frac{v_2}{v_1} = 2.$$

Conserving energy of the system

$$-\frac{GMm}{4R} + \frac{1}{2}mv_1^2 = -\frac{GMm}{R} + \frac{1}{2}mv_2^2$$

$$\frac{1}{2}v_2^2 - \frac{1}{2}v_1^2 = \frac{3}{4} \frac{GM}{R} \Rightarrow v_1^2 = \frac{1}{2} \frac{GM}{R};$$

$$v_1 = \frac{1}{\sqrt{2}} \sqrt{64 \times 10^6} = \frac{8000}{\sqrt{2}} \text{ m/s} \Rightarrow X = 8$$

3. **Ans (5)**

$f_{\text{observer for source 'A'}}$

$$= f_0 \left[\frac{V_{\text{sound}} - V_{\text{medium}}}{V_{\text{sound}} - V_{\text{medium}} + V_{\text{source}}} \right] = \frac{33}{34} f_0$$

$f_{\text{observer for source 'B'}}$

$$= f_0 \left[\frac{V_{\text{sound}} + V_{\text{medium}}}{V_{\text{sound}} + V_{\text{medium}} - V_{\text{source}}} \right] = \frac{35}{34} f_0$$

$\therefore$ Beat frequency

$$= f_1 - f_2 = \left(\frac{35 - 33}{34} \right) f_0 = 5$$

4. **Ans (2)**

At point O

$$\mu_1 \sin i = \mu_2 \sin r$$

$$\frac{3}{2} \times \sin \theta_c = \frac{4}{3} \sin r$$

$$\sin r = \frac{3}{4}$$

At point E

$$\frac{4}{3} \times \frac{3}{4} = 1 \times \sin \theta$$

$$\sin \theta = \frac{\pi}{2}$$

5. Ans (6)

For any small change of pressure dp , there will be a change of volume dV and $dp = -B \frac{dV}{V}$.

In this change, work is done on the system and the energy stored in the material is

$$dW = -pdV \left(\frac{V}{B} \right) dp$$

In the change mentioned in the question, the total work done is

$$W = - \int_V^V pdV = \int_V^V \frac{V}{B} dp$$

The change in volume is negligible and volume can be treated as constant.

$$W = \frac{V_0}{B} \int_{p_0}^p pdp = \frac{V_0}{B} \left(\frac{p^2 - p_0^2}{2} \right)$$

Extra energy stored per unit volume is

$$\frac{W}{V_0} = \frac{1}{2B} (p^2 - p_0^2)$$

CHEMISTRY

SEC-1

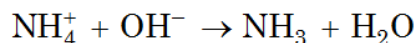
4. Ans (B)

$$5 = pK_B + \text{LOG} \frac{[\text{NH}_4^+]}{0.25}$$

$$[\text{NH}_4^+] = 0.5$$

$$[\text{NH}_4^+] = 0.25$$

m moles of KOH added = 50

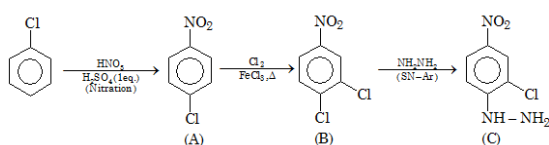


100	50	50
50	0	100

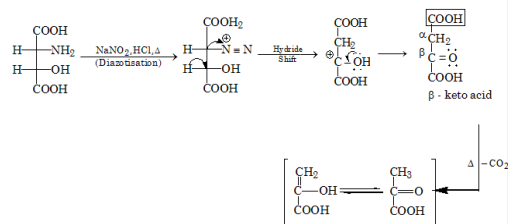
$$p\text{OH} = pK_b + \log \frac{50}{100}$$

$$p\text{OH} = 4.7 - 0.3 \Rightarrow 4.4$$

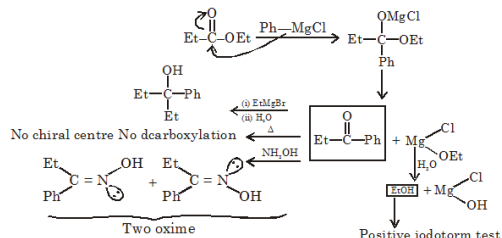
5. Ans (A)



6. Ans (D)



7. Ans (D)



8. Ans (C)

Fake diamonds can be identified by using thermal conductivity.

10. Ans (B)

Since calcium and magnesium salts are soluble, the detergent ion formed will give lather.

CHEMISTRY

SEC-2

4. Ans (A,B,D)

With increase in % s-character the size of hybrid orbital decreases. So bond length decreases, bond energy increases electronegativity increases. Bond angle increases from sp^3 to sp^2 to sp also geometry changes Tetrahedral to planar triangular or linear.

5. Ans (A,B,C,D)

- (A) CN^- is stronger ligand than NH_3
- (B) en is bidentate ligand cannot stretch to 180° to form chelate complex
- (C) Due to formation of complex.
- (D) Solubility product of AgI is less than AgCl .

CHEMISTRY

SEC-3

2. Ans (4)

$$E = E^\circ - 0.06 \text{ pH}$$

$$0.22 = 0.46 - 0.06 \text{ pH}$$

$$0.06 \text{ pH} = 0.46 - 0.22$$

4. Ans (9)

$$\text{Apply } 16(X_A - X_B) + 3.5(X_A - X_B)_2$$

MATHEMATICS

SEC-1

1. **Ans (D)**

The given expression can be written as

$$(1+x)^n(1+y)^n \left(1 + \frac{1}{xy}\right)^n \text{ and constant term is } C_0^3 + C_1^3 + C_2^3 + \dots + C_n^3$$

2. **Ans (D)**

$$\frac{dy}{dx} = \frac{x^2}{4+x^2} \text{ and } \cot\left(\frac{y-x}{2}\right) = \frac{x}{2}$$

$$I = \int \frac{\left(\frac{x^2}{4+x^2}\right) dx}{\left(\frac{x^2-4}{x^2+4}\right) \sin x + \frac{4x}{x^2+4} \cos x}$$

$$= \int \frac{dy}{\cos(y-x) \sin x + \sin(y-x) \cos x} = \int \frac{dy}{\sin y}$$

3. **Ans (B)**

$$\cos \theta = \frac{OA^2 + OB^2 - AB^2}{2OA \cdot OB}$$

$$= \frac{OA^2 + OB^2 - \left(\frac{OA+OB}{2}\right)^2}{2OA \cdot OB}$$

$$= \frac{3}{8} \frac{(OA^2 + OB^2)}{OA \cdot OB} - \frac{1}{4}$$

For maximum $\cos \theta$,

$$\frac{3}{8} \left(\frac{OA^2 + OB^2}{OA \cdot OB}\right) - \frac{1}{2} \geq \frac{3}{8} \times \frac{2OA \cdot OB}{OA \cdot OB} - \frac{1}{4} = \frac{1}{2}$$

$$\Rightarrow \theta = \frac{\pi}{3}$$

4. **Ans (D)**

Let no. of rows in a triangle = k

$$1 + 2 + 3 + \dots + k = n$$

$$\Rightarrow n = \frac{k(k+1)}{2}$$

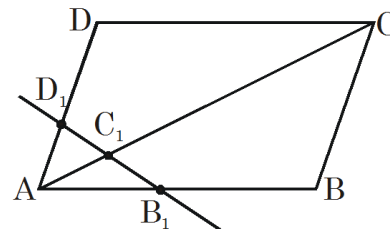
$$\text{Also } \frac{k(k+1)}{2} + 49 = (k+3)^2 \Rightarrow k = 5$$

5. **Ans (A)**

$$\vec{AB}_1 = \lambda_1 \vec{AB}$$

$$\vec{AD}_1 = \lambda_2 \vec{AD}$$

$$\vec{AC}_1 = \lambda_3 \vec{AC}$$



As, ABCD is a parallelogram

$$\Rightarrow \vec{AB} + \vec{BC} = \vec{AC}$$

$$\Rightarrow \vec{AB} + \vec{AD} = \vec{AC}$$

$$\Rightarrow \frac{\vec{AB}_1}{\lambda_1} + \frac{\vec{AD}_1}{\lambda_2} = \frac{\vec{AC}_1}{\lambda_3}$$

$$\Rightarrow \frac{\vec{AB}_1}{\lambda_1} + \frac{1}{\lambda_2} \vec{AD}_1 + \left(\frac{-1}{\lambda_3}\right) \vec{AC}_1 = \vec{0}$$

Here points B, C and D are collinear

$$\Rightarrow \frac{1}{\lambda_1} + \frac{1}{\lambda_2} + \frac{-1}{\lambda_3} = 0 \Rightarrow \frac{1}{\lambda_3} = \frac{1}{\lambda_1} + \frac{1}{\lambda_2}$$

6. **Ans (B)**

If $[x^3 + x^2 + x + 1] = [x^3 + x^2 + 1] + x$, then x is integer

$$\Rightarrow \log|[x]| = 2 - |[x]| \text{ has same solutions as}$$

$$\log|x| = 2 - |x|, x \text{ is integer}$$

$$\Rightarrow \text{No integral solution } \nexists 0$$

7. **Ans (C)**

Let hyperbola is $xy = c^2$ then $\left(ct, \frac{c}{t}\right)$ lies on circle

$$\therefore c^2 t^4 - 2ct^3 - 20t^2 - 4ct + c^2 = 0$$

$$\text{then } \sum t_1 = \frac{2}{c}, \sum t_1 t_2 = \frac{-20}{c^2},$$

$$\sum t_1 t_2 t_3 = \frac{4}{c}, t_1 t_2 t_3 t_4 = 1$$

$$\therefore \ell = 2, m = 44, n = 56$$

8. Ans (B)

$$Q = \sum_{r=0}^n \frac{\sin 3^r \theta}{\cos 3^{r+1} \theta}$$

$$= \frac{\sin \theta}{\cos 3\theta} + \frac{\sin 3\theta}{\cos 9\theta} + \frac{\sin 9\theta}{\cos 27\theta} + \dots + \frac{\sin 3^n \theta}{\cos 3^{n+1} \theta}$$

Now as

$$\frac{\sin \theta}{\cos 3\theta} - \frac{2 \sin \theta \cos \theta}{2 \cos \theta \cos 3\theta} = \frac{\sin 2\theta}{2 \cos \theta \cos 3\theta}$$

$$= \frac{1}{2} \left[\frac{\sin(3\theta - \theta)}{\cos \theta \cos 3\theta} \right]$$

$$= \frac{1}{2} \left[\frac{\sin 3\theta \cos \theta - \cos 3\theta \sin \theta}{\cos \theta \cos 3\theta} \right]$$

$$= \frac{1}{2} [\tan 3\theta - \tan \theta]$$

∴

$$Q = \frac{1}{2} [\tan 3\theta - \tan \theta + \tan 9\theta - \tan 3\theta + \dots + \tan 3^{n+1} \theta - \tan 3^n \theta]$$

$$= \frac{1}{2} [\tan 3^{n+1} \theta - \tan \theta]$$

$$\theta = \frac{P}{2} \quad \therefore P = 2Q$$

9. Ans (A)

$$P = (-4, 0) \quad Q = (0, 6)$$

Let A = (x, y)

$$\frac{PA}{QA} = \frac{2}{3} \Rightarrow 9PA^2 = 4QA^2$$

$$9[(x+4)^2 + y^2] = 4[x^2 + (y-6)^2]$$

$$5x^2 + 5y^2 + 72x + 48y = 0$$

is equation of circum circle of ΔABC

$$\text{Circumcentre of } \Delta ABC = S = \left(-\frac{36}{5}, -\frac{24}{5} \right)$$

and A is orthocenter

$$SG : GA = 1 : 2$$

⇒

$$SG = \frac{1}{3} SA = \frac{1}{3} \sqrt{\left(\frac{36}{5} \right)^2 + \left(\frac{24}{5} \right)^2} = \frac{4\sqrt{13}}{5} \text{ units}$$

10. Ans (A)

Locus of 'Q' is the line of intersection of the plane $x + 2y + 3z = 4$ and

$$1(x-1) + 1(y-1) + 1(z-1) = 0$$

$$\Rightarrow \text{the line is } \frac{x}{1} = \frac{y-5}{-2} = \frac{z+2}{1}$$

MATHEMATICS

SEC-2

1. Ans (A,B,C)

$$\angle BOC = 2A$$

$$\text{In } \Delta ABE, \angle ABE = \frac{\pi}{2} - A \quad \angle AOM = \pi - 2A$$

$$\frac{c}{b} = e^{i2A} \quad \frac{a}{m} = e^{i(\pi-2A)}$$

$$\frac{ac}{bm} = e^{i\pi} = -1$$

$$\therefore m = \frac{-ac}{b}$$

Similarly

$$\ell = \frac{-bc}{a} \quad n = \frac{-ab}{c}$$

2. Ans (A,B,C,D)

Putting $y = x$ in given functional equation

$$f(x) + f(x) = \frac{1}{x} + \frac{1}{x} \Rightarrow f(x) = \frac{1}{x}$$

$$\int_2^3 \frac{3(f(x))^5 - f(x)}{1 - (f(x))^4} dx = \int_2^3 \frac{\left(\frac{3}{x^5} - \frac{1}{x} \right)}{\left(1 - \frac{1}{x^4} \right)} dx = \int_2^3 \left(\frac{\frac{3}{x^7} - \frac{1}{x^3}}{\frac{1}{x^2} - \frac{1}{x^6}} \right) dx$$

$$\text{Let } \frac{1}{x^2} - \frac{1}{x^6} = t \Rightarrow \left(\frac{6}{x^7} - \frac{2}{x^3} \right) dx = dt$$

$$\text{When } x = 2, t = \frac{1}{4} - \frac{1}{64} = \frac{15}{64}$$

$$x = 3, t = \frac{1}{9} - \frac{1}{729} = \frac{80}{729}$$

∴ Integral

$$\frac{1}{2} \int_{15/64}^{80/729} \frac{dt}{t} = \frac{1}{2} (\ln t)_{15/64}^{80/729} = \frac{1}{2} \ln \left(\frac{2^{10}}{3^7} \right) = \frac{1}{2} \log \frac{2^\alpha}{3^\beta}$$

$$\alpha = 10, \beta = 7$$

3. **Ans (A,B,C)**

$$P(A) = \frac{3}{10}, P(B) = \frac{1}{2}, P(C) = \frac{2}{5}$$

$$P(\text{exactly one})$$

$$= P(A) + P(B) + P(C) - 2P(A \cap B) - 2P(B \cap C) - 2P(C \cap A) + 3P(A \cap B \cap C)$$

$$= \frac{3}{10} + \frac{1}{2} + \frac{2}{5} - 2 \times \frac{3}{10} \times \frac{1}{2} - 2 \times \frac{1}{2} \times \frac{2}{5} - 2 \times \frac{2}{5} \times \frac{3}{10} + 3 \times \frac{3}{10} \times \frac{1}{2} \times \frac{2}{5}$$

$$= \frac{3}{10} + \frac{5}{10} + \frac{4}{10} - \frac{3}{10} - \frac{4}{10} - \frac{12}{50} + \frac{9}{50} = \frac{5}{10} - \frac{3}{10} + 3 \times \frac{3}{10} \times \frac{1}{2} \times \frac{2}{5}$$

P(only A passing the exam)

=

$$P(A) P(\bar{B}) P(\bar{C}) = \frac{3}{10} \times \frac{1}{2} \times \frac{3}{5} = \frac{9}{100}$$

$$\therefore \text{Required probability} = \frac{\frac{9}{100}}{\frac{44}{100}} = \frac{9}{44} = \frac{m}{n}$$

4. **Ans (A,B,D)**

Let two perpendicular chords through $A(x_1, y_1)$ be PQ and RS

$$\text{Equation of PQ is } \frac{x - x_1}{\cos \theta} = \frac{y - y_1}{\sin \theta} = r$$

Where $\tan \theta$ = slope of PQ

An point on this line may be taken as

$$(x_1 + r \cos \theta, y_1 + r \sin \theta)$$

As the point lies on $y^2 = 4ax$,

$$\therefore (y_1 + r \sin \theta)^2 = 4a(x_1 + r \cos \theta)$$

$$\Rightarrow r^2 \sin^2 \theta + 2y_1 \sin \theta r - 4a \cos \theta r + y_1^2 - 4ax_1 = 0$$

$$\therefore r_1 r_2 = \frac{(y_1^2 - 4ax_1)}{\sin^2 \theta}$$

$$\text{Similarly } r_3 r_4 = \frac{(y_1^2 - 4ax_1)}{\sin^2 \alpha}$$

where $\tan \alpha$ = slope of RS

Since PQ is perpendicular to RS,

$$\therefore \alpha = 90^\circ + \theta \text{ or } \theta - 90^\circ$$

In either case ; $\sin^2 \alpha = \cos^2 \theta$

Now

$$\frac{1}{r_1 r_2} + \frac{1}{r_3 r_4} = \frac{\sin^2 \theta + \cos^2 \theta}{(y_1^2 - 4ax_1)} = \frac{1}{(y_1^2 - 4ax_1)}$$

which is constant

5. **Ans (A,B)**

$f(x)$ is non differentiability at

$$x = 1, 4, 9, \dots, 2401, 2500$$

$$\therefore m = 50$$

$F(x)$ has minimum value at all x

in $[625, 676]$

$$\therefore \text{Number of integral values of } x \text{ i.e. } n = 52$$

$$\text{Hence, } m + n = 50 + 52 = 102$$

MATHEMATICS

SEC-3

1. **Ans (8)**

$$\text{Let } \vec{a} = \hat{i} + 2\hat{j} + 2\hat{k},$$

$$\vec{b} = (\cos x \cos y) \hat{i} + (\sin y) \hat{j} + (\sin x \cos y) \hat{k}$$

$$\text{here } |\vec{a}| = 3 \text{ and } |\vec{b}| = 1$$

$\therefore \vec{a}$ and $\vec{b}$ are collinear.

$$\Rightarrow \tan x = 2 \text{ and } \tan y = 2 \cos x$$

$$\Rightarrow \tan^2 y = \frac{4}{\sec^2 x} + \frac{4}{1 + 4} = \frac{4}{5}$$

2. **Ans (3)**

$$\text{Given } f(3x) = f(x)$$

$$\therefore f(3x) = f(x) = f\left(\frac{x}{3}\right) = \dots = f\left(\frac{x}{3^n}\right)$$

$$\text{as } n \rightarrow \infty; 3n \rightarrow \infty$$

$$\therefore f(3x) = f(0) \Rightarrow f(x) \text{ is a constant function}$$

$$\therefore f(x) = 5 \quad \forall x \in \mathbb{R}$$

$$\text{sgn}(f(x)) = 1$$

3. **Ans (5)**

$$175 = 5^2 \cdot 7 \cdot 245 = 5 \cdot 7^2 \cdot 875 = 5^3 \cdot 7 \cdot 1715 = 5 \cdot 7^3$$

$$\text{Let } \alpha = \log 5, \beta = \log 7$$

$$a = \frac{\log 175}{\log 245} = \frac{2\alpha + \beta}{\alpha + 2\beta}$$

$$b = \frac{\log 875}{\log 1715} = \frac{3\alpha + \beta}{\alpha + 3\beta}$$

$$\frac{1 - ab}{a - b} = \frac{(\alpha + 2\beta)(\alpha + 3\beta) - (2\alpha + \beta)(3\alpha + \beta)}{(2\alpha + \beta)(\alpha + 3\beta) - (\alpha + 2\beta)(3\alpha + \beta)}$$

$$= \frac{5(\beta^2 - \alpha^2)}{\beta^2 - \alpha^2} = 5$$

4. **Ans (3)**

$$\begin{aligned}
 & \lim_{y \rightarrow \infty} [(8y+3)(y+1)(y+2)(y+3)(2y+3)(4y+5)(2y+7)]^{\frac{1}{7}} - 2y \\
 &= \lim_{y \rightarrow \infty} 2y \left[\left(1 + \frac{3}{8y}\right) \left(1 + \frac{1}{y}\right) \left(1 + \frac{2}{y}\right) \left(1 + \frac{3}{y}\right) \left(1 + \frac{3}{2y}\right) \left(1 + \frac{5}{4y}\right) \left(1 + \frac{7}{2y}\right) \right]^{\frac{1}{7}} - 2y \\
 &= 2 \times \left(\frac{3}{8} + 1 + 2 + 3 + \frac{3}{2} + \frac{5}{4} + \frac{7}{2} \right) \times \frac{1}{7} \\
 &= 2 \times \frac{97}{56} = \frac{194}{56} \Rightarrow \left[\frac{A}{B} \right] = 3
 \end{aligned}$$

 5. **Ans (6)**

$$\frac{x}{1} + \frac{y}{2} + \frac{z}{3} = 1 \quad A(1, 0, 0), B(0, 2, 0), C(0, 0, 3)$$

Foot of perpendicular from (0, 0, 0) the plane of triangle ABC is orthocenter.

Length of perpendicular from origin to the plane

$$6x + 3y + 2z - 6 = 0$$

$$= \frac{6}{7} = k \Rightarrow 7k = 6$$